Keyboards and mice

A computer needs to be fed information before it can process it and then produce an output. Two important ways of inputting information—to interact with the computer directly—are the widely used keyboard and mouse.

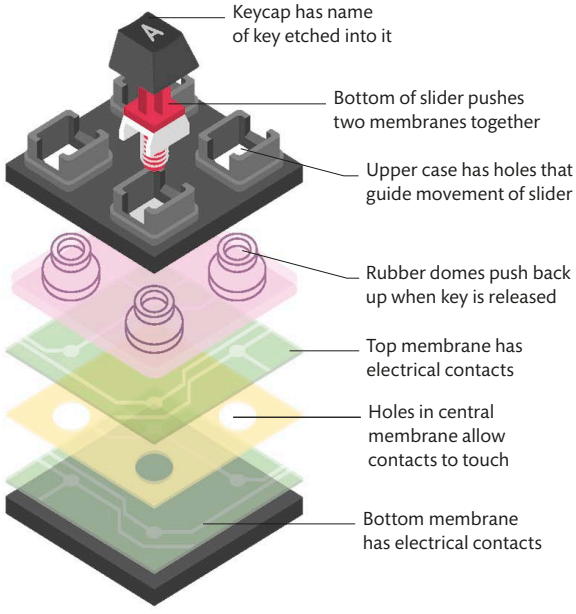
DO I NEED A MOUSE PAD?

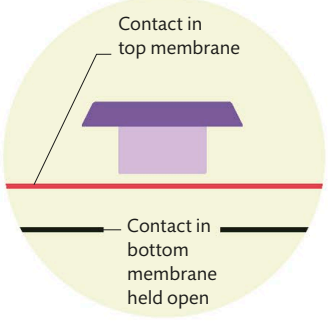
Optical mice do not work well on plain, shiny surfaces, because there is no detail for the mouse's camera to pick up. A mouse pad overcomes this problem.

Keyboard

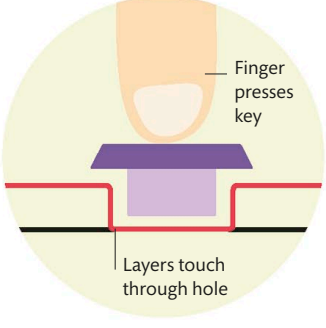
While smartphones and tablets have touch-sensitive keyboards that appear on screen, desktop and laptop computers have keyboards with physical buttons. Inside the keyboard is a number of electric circuits—one for each key. The keys are simple switches, and pressing one completes that key's circuit. The electric current flows to an integrated circuit, which produces a set of binary digits (bits) unique to the key pressed.

Key layers
The most common type of keyboard today uses a technology called "rubber dome over membrane." A slider pushes two contacts together, and a rubber dome provides a force to return the key back to its normal position after it has been pressed.

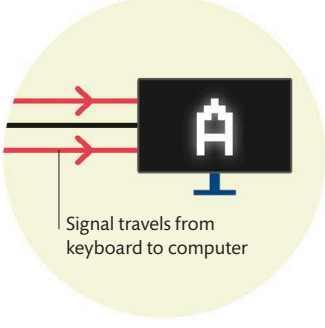




1 Key raised
Beneath each key on a computer keyboard are metal contacts. These contacts are normally held open until they are pressed down.



2 Key pressed down
Pressing the key closes the contacts, allowing electric current to flow through the circuit unique to that key. The current flows to an integrated circuit in the keyboard.



3 Signal sent to computer
The circuit recognizes which key has been pressed and sends a digital signal—a set of binary digits, or scan code—to the computer's main processor.



THE **FASTEST TYPING SPEED** EVER RECORDED WAS **216 WORDS PER MINUTE,** IN 1946